

Zone 3 Methodology Revamp: Trajectory-Trained Scalar Correction

Core proposal: keep Zone 2 physically exact/analytic, and use the neural network only in Zone 3 where the optimal correction is timestep-dependent and has no simple analytic formula.

1. Why the current method needs revision

The current scalar NN was trained to learn the analytic softening correction:

$c_{\text{analytic}} = (r_{\text{soft}} / r)^3$. Multiplying this by the softened force restores the force almost exactly back to Newtonian gravity. This is useful compared with leaving the force softened, but it is not a strong justification for a neural network because the same correction can be computed directly from a formula.

The weakness is therefore methodological, not experimental: the existing NN improves the no-NN softened baseline, but its target is analytically known.

2. Zone 3 is the right place for a real NN contribution

Zone 3 is the macro-step close-encounter region:

$0.05 \text{ AU} < r < 0.15 \text{ AU}$. Here the NN is active, but adaptive sub-stepping is not active. Softening is essentially negligible in this range, so c_{analytic} is approximately 1.0. The remaining error is not a softened-force error; it is a timestep/discretisation error from leapfrog taking a coarse step through a close approach.

This is exactly where a learned correction is defensible: the optimal c is not known from the softening formula and depends on dt , mass scale, and encounter geometry.

3. Revised Zone 3 training target

For each synthetic Zone 3 encounter state, run `ias15` from t to $t + dt$ as the high-accuracy reference. Then run one leapfrog step from the same starting state using different scalar force multipliers c . Choose:

$$c_{\text{opt}} = \operatorname{argmin}_c ||x_{\text{leapfrog}}(c, t+dt) - x_{\text{ias15}}(t+dt)||^2.$$

The NN is then trained to predict $\log(c_{\text{opt}})$ from $[\log(r_{\text{soft}}), \log(m_i), \log(m_j), \log(dt)]$. This changes the learning target from a force-level analytic correction to a trajectory-level timestep-aware correction.

4. Why c can be below 1 in Zone 3

In Zone 3, F_{soft} is already almost Newtonian. However, a coarse leapfrog step can over-bind or over-correct during the encounter. A learned $c < 1$ reduces the one-step impulse so the leapfrog landing position better matches `ias15`. Therefore $c < 1$ is not “weakening Newtonian gravity” as a physical law; it is compensating for finite-timestep impulse error in the numerical update.

5. Why Zone 2 should not use this NN

Zone 2 is handled separately. There, adaptive sub-stepping is active, so the temporal-resolution problem is already addressed. The cleaner approach is to use exact Newtonian force or the analytic softening correction where needed, rather than applying a Zone-3 trajectory correction outside its intended regime.

6. Evidence from the generated data

dt	Median c_{opt} in Zone 3	Interpretation
0.005	0.9988	Tiny correction: leapfrog already resolves the step well
0.040	0.9249	Meaningful impulse reduction at operational dt
0.080	0.8102	Stronger correction as dt becomes coarser
0.100	0.7641	Large dt needs the largest trajectory correction

7. Final mentor-facing position

The revised Zone 3 methodology gives the NN a genuine role: learning a timestep-dependent trajectory correction that the old analytic formula cannot provide. This directly addresses the criticism that the previous scalar NN merely learned $c = (r_{\text{soft}}/r)^3$. The method remains physically constrained because force direction is still analytic, hard fallback remains Newtonian, and the NN only scales the force magnitude in the close-encounter macro-step zone.

Open validation step: after training, the new model should be tested against the current SIMON using full 100-year rollouts, time-averaged RMS, energy drift, and bounded/ejected classification. The one-step data shows the correction is meaningful, but long-horizon benefit must still be verified empirically.